Fiscal Year:	FY 2023	**Task Last Updated:**	FY 04/24/2023
PI Name:	Clement, Gilles Ph.D.		
Project Title:	Neuro-Vestibular Examination During and After Spaceflight (Vestibular Health) (PI: Clement)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-244-5720
Organization Name:	KBR/NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058-3711	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	01/30/2019	**End Date:**	09/30/2033
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Project start date changed 01/30/2019 to match original investigation, as requested by NASA HHC (Ed., 1/3/25). NOTE: End date changed to 09/30/2033 per C. Ribeiro/NASA-JSC. (Ed. 10/5/23). NOTE: Project start date updated due to P.I. change to G. Clement. Information per C. Ribeiro/NASA-JSC. (Ed. 8/29/22).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dervay, Joseph M.D. (NASA Johnson Space Center) Makishima, Tomoko M.D., Ph.D. (University of Texas Medical Branch at Galveston) Wood, Scott Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Adaptation to the absence of Earth's gravitational environment during spaceflight causes neurological disturbances that are either directly or indirectly mediated by the vestibular system. These disturbances include space motion sickness, spatial disorientation, and cognitive impairment, as well as changes in head-eye coordination, vestibulo-ocular reflex, and interactions with support surfaces. After return to Earth, astronauts experience other vestibular-driven behavioral changes, including re-entry motion sickness, motion illusions, gaze-induced nystagmus, and balance and locomotion deficits. Otolith-mediated reflex gain changes are striking shortly after g-transitions. However, animal studies have shown that structural modifications of the vestibular sensory apparatus may occur throughout an extended spaceflight exposure. To date, no long-duration flight studies have directly investigated potential changes in the vestibular organs of astronauts. As a part of the Complement of Integrated Protocols for Human Research (CIPHER), this study will address this gap by performing a systematic neuro-vestibular examination of crewmembers in orbit at regular intervals, as well as immediately after landing. Crewmembers from short-duration, six-month, and one-year missions will be recruited to investigate temporal changes, and to identify trends in adaptation to vestibular health and performance. In orbit, subjects will perform active movements and be exposed to various passive maneuvers executed by the operator. Observations and recordings of eye, head, and body movements, as well as subjective perception of motion and verbal reports, will be used for evaluating the presence of abnormal eye movements, dysmetria, motion sickness symptoms, and illusions of motion during head or body movements. Tests will be performed both early in the mission and once every one or two months thereafter. The post-flight examination will be performed on R+0 and several times thereafter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The tests proposed in this study are well established and validated protocols that are able to detect acute or chronic vestibular syndromes. Repeated measurements during adaptation to g-transitions will provide insight into the vestibular compensation process. If the observed physiological changes in the crewmembers are more deleterious after the year-long International Space Station (ISS) expeditions than those documented after standard-duration ISS expeditions, then relevant countermeasures will be required to enable longer duration missions. Depending on the etiology of the vestibular disorders revealed by our tests, monitoring for long-term health outcomes and vestibular rehabilitation countermeasures can be tailored to the deficits observed.
Task Progress:	This past reporting period, the Neuro-Vestibular Examination During and After Spaceflight (Vestibular Health) study was selected for flight as part of CIPHER's complement of studies (formerly referred to as the integrated One Year Mission project). [Ed. Note: CIPHER is the short title for a set of 14 studies sponsored by NASA and international partner agencies (https://). CIPHER stands for "Complement of Integrated Protocols for Human Exploration Research".] A commercial binocular video eye tracker (Falcon DX goggles, Neurolign Technologies Inc, Toronto, Canada) was qualified for flight and flown to the International Space Station. The ground and inflight protocols were further refined on this platform and custom software changes were implemented to sequence the operator and subject through the experiment protocol using recorded instructions and audio prompts. The inflight and post-flight motion sickness questionnaires were customized to support collaboration across related studies. The Ocular Alignment investigation was added to the pre-flight Vestibular Health study and the investigators (Drs. Mark Shelhamer and Michael Schubert) were added to the investigator team. An Informed Crew Briefing (ICB) was presented to Crew-5 in December 2021, Crew-6 and 69S in March 2022, and Crew-7 and 70S in August 2022. At this point, 2 crewmembers have volunteered to participate in CIPHER. Several inflight operator training sessions have been completed, as well as a few pre-flight baseline data collection sessions. The analysis of the ground-based study performed in 2021 on 31 subjects in the laboratory using the same equipment and procedures as the inflight and ground protocols was completed. Asymmetry was measured – using perceptual reports during unilateral centrifugation, oculomotor responses during visual-alignment tasks, vestibuloocular reflex gain during head-impulse tests, and body rotation during stepping tests. A significant correlation was observed between asymmetries of subjective visual, vertical, and verbal reports during unilateral centrifugation. Another significant correlation was observed between the asymmetries of ocular alignment, vestibulo-ocular reflex gain, and body rotation. These ground-based data suggest that there are underlying vestibular asymmetries in healthy individuals that are consistent across various vestibular challenges. In addition, these findings have value in guiding test selection during experimental design for assessing vestibular asymmetry in healthy adults. The results of this control study were published in a peer-reviewed article and presented at the NASA Human Research Program Investigators' Workshops in 2022 and 2023.
Bibliography Type:	Description: (Last Updated: 10/10/2025)
Abstracts for Journals and Proceedings	Macaulay T, Wood SJ, Reschke MF, De Dios Y, Dervay JP, Makishima T, Schubert MC, Shelhamer M, Clément G. "Neuro-vestibular examination during and following spaceflight (Vestibular Health)." 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. , Feb-2023
Articles in Peer-reviewed Journals	Macaulay TR, Wood SJ, Bollinger A, Schubert MC, Shelhamer M, Bishop M, Reschke MF, Clément G. "Comparison of asymmetry between perceptual, ocular, and postural vestibular screening tests. " Brain Sci. 2023 Jan 23;13(2):189. https://doi.org/10.3390/brainsci13020189 ; PMID: 36831732; PMCID: PMC9954277 , Jan-2023
Books/Book Chapters	Clément G, Wood S. "Space physiology " in "Primer on the Autonomic Nervous System (Fourth Edition)." Ed. I Biaggioni, K Browning, G Fink, J Jordan, PA Low, JFR Paton JFR. Academic Press, 2023. p. 329-32. https://doi.org/10.1016/B978-0-323-85492-4.00058-2 , Jan-2023